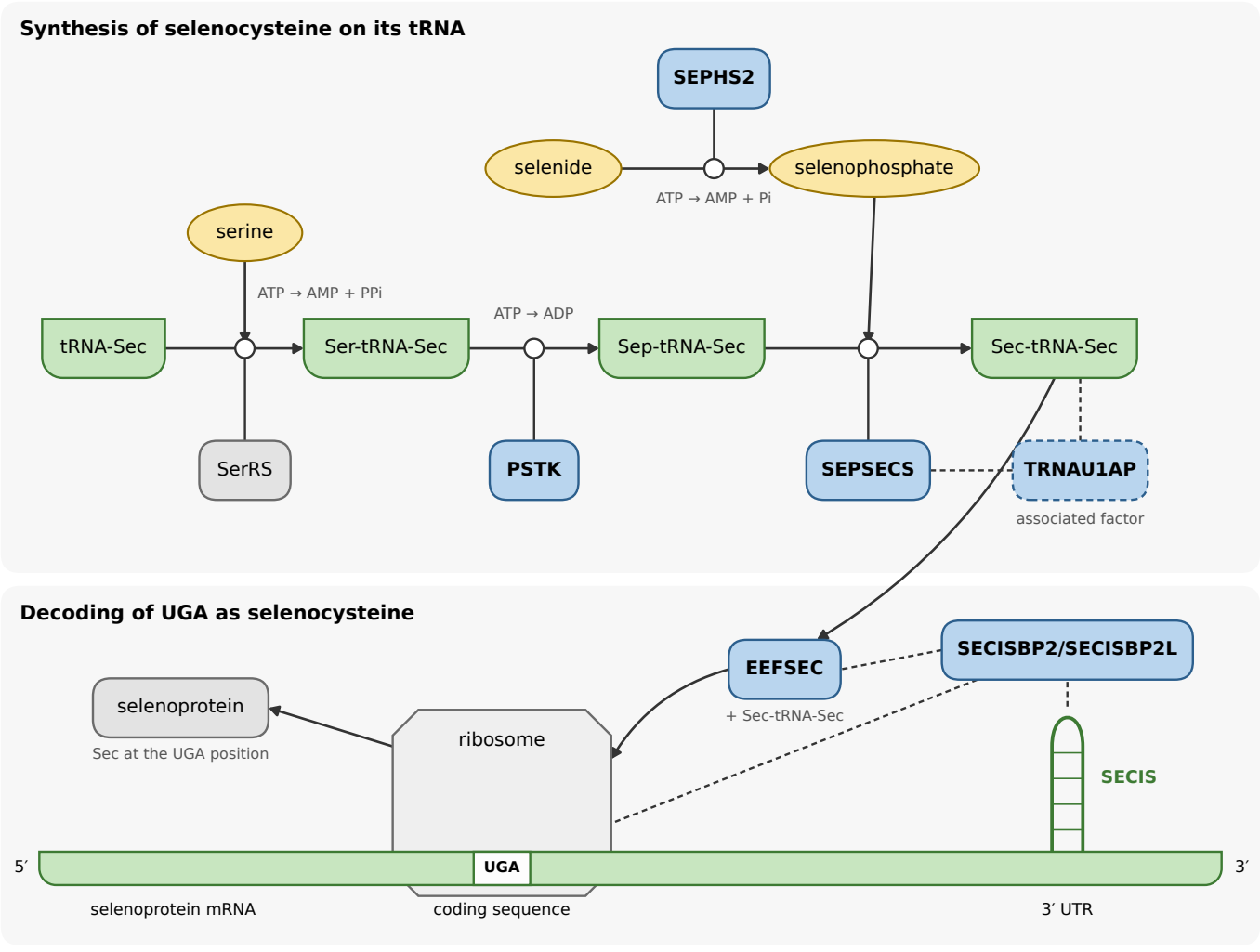
Synthesis of selenocysteine on its tRNA
SEPHS2
selenide
selenophosphate
ATP → AMP + Pi
serine
ATP → AMP + PPi
ATP → ADP
tRNA-Sec
Ser-tRNA-Sec
Sep-tRNA-Sec
Sec-tRNA-Sec
SerRS
PSTK
SEPSECS
TRNAU1AP
associated factor
Decoding of UGA as selenocysteine
EEFSEC
+ Sec-tRNA-Sec
SECISBP2/SECISBP2L
selenoprotein
Sec at the UGA position
ribosome
SECIS
5′
UGA
3′
selenoprotein mRNA
coding sequence
3′ UTR
metabolite
RNA
protein in Table MACHINERY
other protein
associated factor
complex
conversion, transfer
catalysis
binding